Job Ladders, Human Capital, and the Cost of Job Loss*

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Abstract

High-tenure workers who lose their jobs experience a large and prolonged fall in wages and earnings. To quantify the forces behind this empirical regularity, we propose a rich structural model of the labor market with heterogeneous firms, on-the-job search, and specific and general human capital. By jointly matching moments of workers' mobility and wages, the model can replicate the losses in earnings and wages observed in the data. The loss of specific human capital is the primary driver of the cumulated wage losses post-displacement (49%), followed by the loss of a job with a more productive employer (33%).

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1 Introduction

A large body of empirical research has established the existence of large and persistent earnings losses following job displacement for high-tenure workers. For example, Davis and von Wachter (2011) find that, in the United States, displaced male workers with more than three years of tenure lose the equivalent of 12% of the present value of earnings in the absence of displacement. Schmieder, von Wachter and Heining (2023) estimate even larger losses of 15% for Germany. The aim of this paper is to quantify the drivers behind this empirical regularity using a rich structural model of the labor market.

Workhorse search models of the labor market with on-the-job search and firm heterogeneity imply that earnings losses reflect the loss of a good job (e.g., Burdett and Mortensen, 1998; Postel-Vinay and Robin, 2002). These models feature a *job ladder* that workers climb, moving toward higher-paying jobs, over the course of their career. The positive association between employment tenure and wages (and therefore the large drop in earnings after a displacement event) reflects the fact that workers keep searching for better employers until they settle in high-productivity jobs, which both pay more and last longer.

Another view with a long tradition in labor economics is that the positive association between tenure and earnings losses reflects the accumulation of skills that are productive (and therefore reflected in wages) only with the current employer but not with future employers (e.g., Becker, 1964; Topel, 1990; Lazear, 2009). Human capital is, to some degree, *specific.*[1] In this framework, earnings losses reflect the loss of employer-specific skills that are accumulated with tenure.[2]

Finally, workers may accumulate general skills while employed and these may deteriorate when nonemployed (e.g., Ljungqvist and Sargent, 1998). Therefore, earnings losses may also reflect lower accumulation of general human capital for displaced workers relative to a counterfactual path in the absence of displacement.

In this paper, we provide a unifying framework featuring all three of these mechanisms and use it to quantify their relative contribution to the long-run losses in wages and earnings experienced by displaced workers. We build and estimate a structural search model of the labor market with the following key ingredients: heterogeneous firms, on-the-job search,

[1]While Becker's (1964) distinction between general (i.e. fully portable) and specific capital is very stark in the perfectly competitive environment he considered, Acemoglu and Pischke (1999) have forcefully argued that, in the presence of frictions, specificity actually captures the fraction of total human capital that is lost *in expected value* upon changing job.

[2]Learning about match quality, as in Jovanovic (1979), can also generate a positive association between wages and tenure on the job. We abstract from this channel. As discussed in Nagypál (2007), it is difficult to distinguish learning about imperfectly observed productivity from actual productivity dynamics on the basis of wage and spell data.

specific and general human capital accumulation, and endogenous job loss.

The model is estimated on longitudinal German Social Security data using indirect inference. It can reproduce the size and persistence of the post-displacement earnings and wage losses observed in the data. We use the estimated model to decompose the post-displacement wage losses into three components: (i) job ladder losses (displaced workers are re-employed at firms that are, on average, less productive than their previous employer); (ii) general skill losses (displaced workers do not accumulate general human capital while nonemployed); and (iii) losses of specific skills (displaced workers lose human capital accumulated with the last employer, which is not productive with other employers). We find that the loss of specific human capital accounts for a significant fraction (46–52%) of cumulative wage losses for displaced workers. Although job ladder losses are still an important driver, their contribution is significantly lower (30–36%) than implied by studies that do not take into account specific capital, such as Krolikowski (2017), Jung and Kuhn (2019), Burdett, Carrillo-Tudela and Coles (2020), and Jarosch (2023).

In the model, both nonemployed and employed workers sample job offers infrequently from an exogenous firm productivity distribution. Nonemployed workers have a lower reservation productivity than employed workers, but they climb the job ladder by accepting subsequent offers from more productive employers while employed. Wages are set according to the sequential auction negotiation protocol in Postel-Vinay and Turon (2010), where wages are renegotiated only if either party has a credible threat: wages respond both to outside offers as in Cahuc, Postel-Vinay and Robin (2006) and to match-specific shocks. Employed workers accumulate general human capital, which is transferable to other employers, but may depreciate during nonemployment. They also accumulate specific skills, which, in contrast, are only valuable with their current employer. The model features endogenous job destruction. As they climb the job ladder, workers sort into more productive jobs, which are also more stable, since they are less likely to be destroyed following negative productivity shocks.

In this framework, displaced high-tenure workers lose a job with a more productive employer, as well as the specific skills associated with that job. Besides, their general skills also depreciate during nonemployment, further reducing their productivity when re-employed. Upon re-employment, they are more likely to accept a low productivity job that, by also being less stable, does not favor the acquisition of general and specific skills, further hindering the recovery of earnings and wages.

The model is able to replicate the returns to on-the-job tenure within firms, the returns to experience, as well as the profile by tenure of the job-switching rates observed in the data. Additionally, it delivers large and persistent earnings and wage losses that mimic their counterparts in the data. Similarly to the data, most of the persistence in earnings comes from wages, which drop by more than 10% upon separation and only slowly recover after re-employment.

Related literature The paper is motivated by the large empirical literature documenting large and persistent earnings losses for job losers with high pre-displacement job tenure relative to counterfactual non-separators (Jacobson, LaLonde and Sullivan, 1993; Couch and Placzek, 2010; Davis and von Wachter, 2011; Flaaen, Shapiro and Sorkin, 2019). In particular, a recent literature finds that declines in wages account for a large share of long-term earnings losses and builds on Abowd, Kramarz and Margolis's (1999) (AKM) model to quantify the contribution of employer-specific wage premia (the firm effects in the AKM decomposition) to long-term wage losses. For Germany, Schmieder et al. (2023) and Fackler, Müller and Stegmaier (2021) find that employer effects account for approximately 90% of long-term wage losses. By contrast, Lachowska, Mas and Woodbury (2020) find that employer effects account for only 17% of wage losses for workers displaced in the state of Washington during the Great Recession.[3] Moore and Scott-Clayton (2025) estimate a higher share (24%) for workers displaced in Ohio during the relatively tight labor market of 2002–2008. Bertheau, Acabbi, Barceló, Gulyas, Lombardi and Saggio (2023) apply the same harmonized design to seven European countries over a common sample period and find that employer premia account for shares of wage losses ranging from 35% in Spain to 100% in Portugal.[4] All in all, it seems safe to conclude that the range of estimates in the literature stems both from differences in sample periods *and* cross-country institutional differences, such as the relative prevalence of collective bargaining agreements.

The paper also builds on models of wage and employment dynamics in the presence of labor market frictions (e.g., Yamaguchi, 2010; Postel-Vinay and Turon, 2010; Bagger, Fontaine, Postel-Vinay and Robin, 2014). It is related to a growing quantitative literature that aims to understand the extent to which the empirical evidence on the costs of job loss can be explained within these models. The idea of modeling a job ladder in terms of firm productivity with on-the-job search and endogenous separation first appears in Krolikowski (2017) and Jung and Kuhn (2019). The resulting job ladder implies that firm productivity, wages, and job security increase with employment tenure. Since job and employment tenure

[3] Schmieder et al. (2023) show that the contrasting findings are partly explained by compositional effects, namely differences in transition rates across the rank distribution of employer effects in the two countries. Reweighting Lachowska et al.'s (2020) wage and employer-effects losses using the German transition rates approximately doubles the share of wage losses explained.

[4] Raposo, Portugal and Carneiro (2021) estimate a significantly lower share (31%) for Portugal when controlling for job titles, defined as the interaction of occupation, collective agreement, and salary-scale point within a collective agreement–occupation pair. They find that job titles account for 37% of wage losses in Portugal.

are positively correlated, displaced workers with high job tenure experience large wage losses relative to the counterfactual. Crucially, the persistence of these losses is driven by differences in job destruction rates between job losers and counterfactual job keepers. The former, who on average find re-employment at lower-productivity firms, are more exposed to the risk of becoming non-employed and repeatedly falling back to the bottom of the job ladder. Conversely, the latter remain employed at stable, high-productivity firms. Both papers show that this mechanism can account for the large and persistent wage and earnings losses estimated for the United States. Skill accumulation does not feature in Krolikowski (2017), while Jung and Kuhn (2019) allow for general human capital (experience) accumulation on the job but do not explicitly quantify its contribution to wage and earnings losses.[5] Huckfeldt (2022) shows how endogenously selective hiring can account for the cyclical behavior of the present value of earnings losses from job loss documented by Davis and von Wachter (2011).

Two important contributions estimate structural models of displacement losses using the same German administrative data as this paper. Burdett, Carrillo-Tudela and Coles (2020) show that forgone general human capital accumulation during nonemployment can account for the size and persistence of these losses, even in the absence of heterogeneity in employer productivity or job displacement rates. Their model augments Burdett and Mortensen (1998) with workers' risk aversion and on-the-job accumulation of general human capital. The equilibrium features a non-degenerate distribution of contracts offering different increasing wage–tenure profiles, but all wages are proportional to workers' current stock of general human capital. The latter is the main driver of *long-term* wage losses, as displaced workers forgo experience accumulation while unemployed relative to counterfactual job stayers. By contrast, losses due to falling down the job ladder in the wage-tenure offer distribution are short-lived in the calibrated model. They converge to zero within six months for workers with a high-school degree and no vocational training, and within twelve months for workers with vocational training. The difference reflects the higher rate of job-to-job transitions among the latter group.

Jarosch (2023) is the paper closest to ours. It combines the key features of Jung and Kuhn (2019) and Burdett et al. (2020) to jointly study the contributions of employer heterogeneity, both in productivity and (exogenous) job destruction, and forgone general human capital accumulation during nonemployment to wage and earnings losses. Jarosch (2023) imposes the negative correlation between firm effects and level job destruction estimated from the data and finds that the loss of a productive and secure job is the main driver, accounting for about

[5] The contribution of general human capital to wage losses is included in their estimated selection effect, relative to earlier estimates of job losses (e.g., Jacobson et al., 1993; Couch and Placzek, 2010), which impose that the control group of non-displaced workers is continuously employed.

70% of wage losses over the first ten years following displacement. Forgone skill accumulation accounts for the remainder and largely reflects the loss of job security associated with falling off the job ladder.

Our point of departure is to allow for the accumulation of specific human capital, along the lines of Becker (1964), Topel (1990), Dustmann and Meghir (2005), and Lazear (2009), as an additional potential source of post-displacement losses and to endogenize level job destruction and, therefore, its negative correlation with firm effects. We can then quantitatively assess the contribution of specific human capital relative to the other mechanisms described above within a unified framework. specific human capital accumulation implies a negative relationship between tenure and separation rates even after controlling for unobserved worker and employer heterogeneity, as well as experience. This is because, all else equal, longer job tenure is associated with higher productivity in the current job. This implication distinguishes our framework from those of Krolikowski (2017), Jung and Kuhn (2019), Burdett et al. (2020), and Jarosch (2023). We show in Section 3.5 that this prediction is consistent with the data.

Outline The model is introduced in Section 2. Section 3 describes the data, the identification strategy, and the estimation results. Section 4 uses the estimated model to decompose the cost of job loss. Finally, Section 5 concludes.

2 Model

2.1 Environment

We model a frictional labor market in which both employed and nonemployed workers search for jobs with heterogeneous productivity. Time is discrete and goes on forever. The economy is populated by risk-neutral workers and firms with common discount factor β.

Firms differ in their observable productivity θ, which is drawn from an exogenous distribution $F(\theta)$ and is constant over time. Workers have stochastic lifetimes and are either employed (labor force status $\ell = E$) or nonemployed (labor force status $\ell = N$).[6] In every period, a fraction κ of the labor force dies and is replaced by an equal mass of ex-ante identical nonemployed new entrants.

[6]There is a tension between the search literature, which typically focuses only on employment and unemployment, and the earnings losses literature, which typically tracks displaced workers regardless of their labor force status. Since wage and earnings losses are our primary focus, we group unemployment and non-participation into a single "nonemployment" state ($\ell = N$), both in the model and in the empirical analysis. As detailed in the empirical part of the paper (Section 3.1), the sample is restricted to workers well-attached to the labor market. See Krolikowski (2017) for a similar approach.

Workers are endowed with general human capital $g \in \{g_0, \dots, g_{n_g}\}$ and specific human capital $s \in \{s_0, \dots, s_{n_s}\}$. Both are initialized at their lowest grid points at the start of a worker's career and job, respectively. While employed, general human capital increases from g_i to $\min\{g_{i+1}, g_{n_g}\}$ with probability ϕ_E; while nonemployed, it falls to $\max\{g_{i-1}, g_0\}$ with probability ϕ_N. specific human capital is entirely lost upon separation. While workers continue in their current match, specific skills rise from s_i to $\min\{s_{i+1}, s_{n_s}\}$ with probability γ.

The labor market is characterized by search frictions. Nonemployed workers get an offer from a potential employer with probability λ_N, while employed workers get an offer from an alternative employer with probability λ_E. Search is random and all workers sample from the common job offer distribution $F(\theta)$.

Once a firm and a worker form a match, the per period output $y(\theta, g, s, \varepsilon)$ is a function of the fixed productivity component θ, the level of general and specific human capital, g and s, and a stochastic productivity shock ε. The initial realization of ε is equal to ε_0 in all new matches, and its subsequent realizations are drawn from a distribution $H(\varepsilon'|\varepsilon)$. As in Mortensen and Pissarides (1994), shocks to match productivity lead to endogenous job destruction events. In particular, when the realization of the shock ε is low enough, the worker and the firm agree to dissolve the match.[7]

Nonemployed workers enjoy flow utility $z(g)$. Indexing flow utility by the level of general human capital g captures the additional resources more experienced workers have access to when nonemployed.[8]

The timing is as follows. At the beginning of the period, a worker dies with probability κ. Surviving workers draw a new value of g, if nonemployed, or a new triplet (g, s, ε), if employed. Following that, nonemployed workers may find a job. Employed workers, instead, are exposed to the following sequence of shocks. First, their job may be destroyed for exogenous economic reasons with probability δ. In such a case, they enter nonemployment with probability $1 - \lambda_R$. With complementary probability λ_R they draw again from the distribution of firm productivity $F(\theta)$ without transiting through nonemployment.[9] Second, workers in surviving

[7]As an alternative modelling strategy, the state variables (s,ε) could be combined in a single match-specific shock with an upward drift. We choose to keep s and ε separate in the model given our emphasis on specific human capital.

[8]Unemployment benefits are typically indexed on workers' previous wages, which are correlated with g in the model. More experienced workers are also likely to have accumulated more liquid assets they can draw on when nonemployed.

[9]This modeling choice implies that not all job-to-job transitions are necessarily the results of an optimal choice, since they may, for example, reflect layoffs announced to workers in advance. When estimating the model, we target the average change in wage following a job-to-job transition to discipline this parameter. Examples of models that feature such relocation shocks include Jolivet, Postel-Vinay and Robin (2006), Bagger et al. (2014), and Bagger and Lentz (2019).

matches may receive an outside offer. This completes the revelation of uncertainty for the current period. The firm and worker optimally decide whether to endogenously end the match or to produce, after possibly renegotiating the current contract wage.

2.2 Wage bargaining

The wage-setting mechanism follows the sequential auction mechanism described in Postel-Vinay and Turon (2010), which is based on the efficient rigid-contract framework of MacLeod and Malcomson (1993). Wages are determined by a fixed-wage contract that is renegotiated when either party has a credible threat. In this framework, wages respond both to outside offers as in Cahuc et al. (2006) and to match-specific shocks. We depart from Postel-Vinay and Turon (2010) by allowing for persistent match-specific shocks and by leaving the bargaining weight parameter unrestricted.

Let $v = (g, s, \varepsilon)$ denote the vector of variables that are subject to shocks. Let $U(g)$ denote the value of nonemployment for a worker with general human capital g and $W(\theta, w, v)$ the value of a worker currently employed at a firm of type θ, being paid the current contract wage w and with residual state vector v. Let $J(\theta, w, v)$ be the corresponding value to the firm of the same filled job. Let

$$S(\theta, v) = \max\{0, W(\theta, w, v) - U(g) + J(\theta, w, v)\} \tag{1}$$

denote the joint surplus, the private net value of the match net of the respective outside options which, given efficient bargaining, is independent of the wage.[10] Finally, we denote by $S_0(\theta, g) = S(\theta, g, s_0, \varepsilon_0)$ the surplus from a newly-formed match between a firm of type θ and a worker with general skills g.

Nonemployed workers If a nonemployed worker with general skills g and a firm of type θ choose to form a match, the initial wage w_0 is such that the worker receives a share $\alpha \in [0, 1]$ of the joint surplus

$$W(\theta, w_0, g, s_0, \varepsilon_0) - U(g) = \alpha S_0(\theta, g). \tag{2}$$

Employed workers without an outside offer Consider an ongoing match with current state (θ, w, v'), where the notation emphasizes that w is the contract wage carried over from the *previous* period and v' is the *current* realization of the vector of shocks. Assuming that continuing the match is jointly efficient, the current contract wage will be renegotiated to a

[10] The firm's outside option is the value of an unfilled job, which is equal to zero.

new value w' if and only if either party can credibly threaten to abandon the match rather than producing at the current contract wage w.[11]

There are three possible cases.

1. If $0 \leq W(\theta, w, v') - U(g') \leq S(\theta, v')$, both parties strictly prefer continuing the match at an unchanged wage rate to their respective outside options. It follows that the contract wage is not renegotiated and $w' = w$.

2. If $W(\theta, w, v') - U(g') < 0 \leq S(\theta, v')$, the worker has a credible threat to quit and enter nonemployment rather than continuing the match at the current wage rate. The firm matches the worker's outside option and the wage is renegotiated to a new value w' satisfying $W(\theta, w', v') = U(g')$.

3. Finally, if $0 \leq S(\theta, v') < W(\theta, w, v') - U(g')$, the firm has a credible threat to end the match and obtain a zero return rather than the negative return $J(\theta, w, v') = S(\theta, v') - (W(\theta, w, v') - U(g'))$ from continuing the match at the current contract wage. In this case, the new wage contract w' gives the firm its outside option and satisfies $W(\theta, w', v') - U(g') = S(\theta, v')$.

Putting these three cases together, the equilibrium value for an employed worker with current state (θ, w, v') and no outside offer is given by

$$\tilde{W}(\theta, w, v') = \max\Big\{U(g'),\ \min\big\{S(\theta, v') + U(g'),\ W(\theta, w, v')\big\}\Big\}, \tag{3}$$

and the transition law $w'(\theta, w, v')$ for the contract wage is implicitly given by

$$W(\theta, w'(\theta, w, v'), v') = \tilde{W}(\theta, w, v'). \tag{4}$$

Employed workers with an outside offer Consider an ongoing match with current state (θ, w, v'). If the worker is contacted by a firm with type $\hat{\theta}$, there are three possible cases, assuming that producing with either the current firm or the outside firm is privately efficient.

1. If $S_0(\hat{\theta}, g') > S(\theta, v')$, the worker switches employers. Switching employers is efficient and the worker's threat point in bargaining with the poaching firm is receiving all the surplus from the current match. The worker's value associated with starting with

[11] In what follows, we omit discussion of the case in which the parties agree to end the match. It follows from the definition of surplus in equation (1) that the payoff formulas we derive apply in this case too and imply zero surplus in the case of separation.

contract wage w_0 at the poaching firm satisfies

$$W(\hat{\theta}, w_0, g', s_0, \varepsilon_0) = U(g') + S(\theta, v') + \alpha\Big[S_0(\hat{\theta}, g') - S(\theta, v')\Big]. \quad (5)$$

2. If $S_0(\hat{\theta}, g') \leq S(\theta, v')$, the worker stays with the current employer, but uses the outside offer to renegotiate the wage up. The threat point in bargaining with the current employer is receiving all the surplus at the poaching firm. The worker's value associated with the renegotiated contract wage w' satisfies

$$W(\theta, w', v') = U(g') + S_0(\hat{\theta}, g') + \alpha\Big[S(\theta, v') - S_0(\hat{\theta}, g')\Big]. \quad (6)$$

   This applies if $W(\theta, w, v') < U(g') + S_0(\hat{\theta}, g') + \alpha\Big[S(\theta, v') - S_0(\hat{\theta}, g')\Big]$.

3. If $S_0(\hat{\theta}, g') \leq S(\theta, v')$, the workers stays with the current employer, but the outside offer is not binding. This applies if $W(\theta, w, v') \geq U(g') + S_0(\hat{\theta}, g') + \alpha\Big[S(\theta, v') - S_0(\hat{\theta}, g')\Big]$. The outside offer does not represent a credible threat and bargaining takes place as if it were not available. The transition law for the contract wage and the associated worker's value satisfy equation (4).

It follows from points 2 and 3 that if a match survives despite the availability of an outside offer, the transition law $w'(\theta, w, v')$ for the contract wage is implicitly given by

$$W(\theta, w'(\theta, w, v'), v') = \max\Big\{\tilde{W}(\theta, w, v'),\ U(g') + S_0(\hat{\theta}, g') + \alpha\big[S(\theta, v') - S_0(\hat{\theta}, g')\big]\Big\}. \quad (7)$$

2.3 Value functions

We now introduce the recursive representation of the agents' problems. Let $\hat{\beta} = \beta(1-\kappa)$ denote the mortality-adjusted discount factor and let $\mathbb{E}_N[\cdot]$ and $\mathbb{E}_E[\cdot]$ denote the expectation operator conditional on the relevant information set, respectively, for nonemployed and employed workers. More explicitly, if $\ell \in \{E, N\}$ denotes employment status, these expectations are given by $\mathbb{E}_N[\cdot] = \mathbb{E}[\cdot|g, \ell = N]$ and $\mathbb{E}_E[\cdot] = \mathbb{E}[\cdot|\theta, w, v, \ell = E]$.

Being nonemployed with general skill level g has value

$$U(g) = z(g) + \hat{\beta}\mathbb{E}_N\left[U(g') + \lambda_N \int \alpha S_0(\hat{\theta}, g') dF(\hat{\theta})\right]. \quad (8)$$

A nonemployed worker has a flow of income, $z(g)$, that depends on her level of general human capital g. At the end of the period, the worker's general skills may be hit by a depreciation shock, after which the worker may find a job that will be accepted as long as it has a positive surplus.

A worker employed in the current period has value

$$
\begin{aligned}
W(\theta, w, v) = w + \hat{\beta}\mathbb{E}_E \Big\{ & \delta\big[U(g') + \lambda_R \int \alpha S_0(\hat{\theta}, g') dF(\hat{\theta})\big] \\
& + (1-\delta)(1-\lambda_E) W(\theta, w', v') + (1-\delta)\lambda_E \int \Big[(1-\mathbb{I}_S) W(\theta, w', v') \\
& + \mathbb{I}_S \big(U(g') + (1-\alpha) S(\theta, v') + \alpha S_0(\hat{\theta}, g')\big)\Big] dF(\hat{\theta}) \Big\}, \quad (9)
\end{aligned}
$$

where $\mathbb{I}_S$ is an indicator function that takes value one if $S_0(\hat{\theta}, g') > S(\theta, v')$ and zero otherwise.

The worker receives the wage w in the current period. At the end of the period, first the shocks to v realize, then the job is exogenously destroyed with probability δ. If the match is not exogenously destroyed, the worker receives no offer with probability $1 - \lambda_E$, in which case the match continues with wage w' given by equation (4). With probability λ_E, instead, the worker receives a job offer. Given (θ, w, v'), the match continues with wage w' given by equation (7) for all values of $\hat{\theta}$ such that the surplus from the current match is larger than the surplus with the poaching firm.[12] Otherwise, the worker switches firms and obtains the value given by equation (5).

Firms Symmetrically, the firm's present value of a job is determined by the asset pricing equation

$$
\begin{aligned}
J(\theta, w, v) = y(\theta, v) - w + \hat{\beta}(1-\delta)(1-\lambda_E)\mathbb{E}_E \max\big\{0, J(\theta, w', v')\big\} \\
+ \hat{\beta}(1-\delta)\lambda_E \mathbb{E}_E \int (1-\mathbb{I}_S) \max\{0, J(\theta, w', v')\} dF(\hat{\theta}).
\end{aligned} \quad (10)
$$

The first term on the right-hand side of equation (10) is the flow of profits in the current period. If the match is viable, the continuation value equals $J(\theta, w', v')$ with w' given by (4) or (7) depending on the availability of an outside job offer to the worker.

Net surplus By combining the expressions for the value of nonemployment (8), the value of employment (9), and the value of a job to the firm (10), together with the definition of

[12]Note that the term $W(\theta, w', v')$ in equation (9) already takes into account the possibility of separation, since it follows from equations (4) and (7) that $W(\theta, w', v') = U(g')$ if the match ends because $S(\theta, w', v') < 0$.

joint worker surplus (1), we obtain the Bellman equation for the latter

$$
\begin{aligned}
S(\theta, v) = \max\Bigg\{ & 0,\ y(\theta, v) - z(g) \\
& - \hat{\beta}\mathbb{E}_N\Big[U(g') + \lambda_N \int \alpha S_0(\hat{\theta}, g') dF(\hat{\theta})\Big] + \hat{\beta}\mathbb{E}_E\Big[U(g') + \delta\lambda_R \int \alpha S_0(\hat{\theta}, g') dF(\hat{\theta})\Big] \\
& + \hat{\beta}(1-\delta)\lambda_E \mathbb{E}_E \int \Big[(1 - \mathbb{I}_S) S(\theta, v') + \mathbb{I}_S\big[(1-\alpha) S(\theta, v') + \alpha S_0(\hat{\theta}, g')\big]\Big] dF(\hat{\theta}) \\
& + \hat{\beta}(1-\delta)(1-\lambda_E)\mathbb{E}_E S(\theta, v')\Bigg\}.
\end{aligned}
\tag{11}
$$

As is standard in models of efficient bargaining with transferable utility, the value of the joint surplus does not depend on the wage. The bargaining protocol only affects how the match surplus is shared between the firm and the worker.

2.4 Model mechanisms

Our model represents the joint evolution of workers' wages and their mobility decisions across labor force states and employers. Here, we highlight the key model mechanisms that contribute to earnings and wage losses for displaced high-tenure workers. We discuss estimation in detail and quantify the importance of these channels in subsequent sections.

As in standard on-the-job search models with offer matching, the model implies that wages grow with experience and tenure (Postel-Vinay and Robin, 2002). Specifically, wages grow with experience both because workers climb the job ladder and because they have had more time to leverage outside offers and extract more of the surplus in their current match. Wages similarly grow with tenure because workers and firms renegotiate wages within the same job spell as workers leverage the relevant offers from other firms. The accumulation of general and specific human capital represents additional channels to account for returns to experience and tenure. Workers can leverage general human capital even in the absence of outside offers as the value of nonemployment $U(g)$ is increasing in g. The accumulation of specific human capital makes the worker surplus larger, which represents additional rents that workers can extract leveraging outside offers.

The model also generates endogenous separations to nonemployment that are decreasing with tenure. This is because high-tenure workers tend to be at high-θ firms and to have accumulated more specific human capital. At a high-θ firm, a negative match-specific shock is less likely to drive the worker surplus below the value of nonemployment. The accumulation of specific skills reinforces this last mechanism since the worker surplus is also increasing in s.

Consider then a group of high-tenure workers losing their job. In order to get back to their pre-separation wage level, they would need to (i) find a similar high-θ employer, (ii) reach a similar level of specific skills s, (iii) compensate for any depreciation in general skills g during nonemployment, and (iv) extract a similar share of the match-surplus by leveraging outside offers. All four mechanisms put downward pressure on wages. They are reinforced by the fact that recently nonemployed workers who find a job are more likely to transition back to nonemployment because these matches have low surplus, which leads to repeated nonemployment spells. This last channel is similar to the ladder in job security described in Jarosch (2023), but it comes from endogenous separations in our setting with match-specific shocks.

3 Quantitative analysis

3.1 Data description and sample selection

This study is based on the Sample of Integrated Labor Market Biographies (SIAB), a matched employer-employee dataset from Germany.[13] The SIAB covers a random 2 percent sample of approximately 1.6 million individuals (excluding civil servants and self-employed workers) registered in the German social security system.

The data contain detailed information on these individuals' employment status (employed, unemployed receiving benefits), type of contract (full-time, part-time), occupation, and (daily) wages. They also include basic demographic information, such as gender, age, and education level. In addition, the data keep track of the establishment identifier in which the worker is employed, along with some general information on its geographic location and sector of activity.

The raw data come as a collection of employment spells for all workers in the sample for each year over the period 1975–2014. We drop all spells that are shorter than a month or with daily wages below ten euros (in 2010 prices), as well as all workers who are not observed for more than a year.[14] We then convert the data from spells to monthly frequency, as described in Appendix A. Any month overlaps are resolved by prioritizing employment over nonemployment, and within employment by keeping the highest-wage full-time job.

We further apply the following sample selection criteria. We focus on male workers between 18 and 64 years old who are only ever employed in West Germany. Since there is no

[13]These data are provided by the Research Data Centre (FDZ) of the German Federal Employment Agency (BA) at the Institute for Employment Research (IAB).

[14]We require that individuals have no gap of more than one year in the German administrative data. Here a gap therefore means a period with no administrative record, where a record can be full-time employment, part-time employment, or receiving benefits.

information on working hours, we restrict the analysis to full-time workers. Because we infer experience and tenure from the observed employment spells, we only retain those workers who can be tracked from the beginning of their career, which is assumed to start shortly after the expected completion date of their studies. Specifically, workers cannot be older than 19 years old if they have no high school diploma when they are first observed in the data. We similarly require that high school graduates cannot be older than 22, graduates from a technical college older than 27, and university graduates older than 27, when they first appear in the dataset.[15] Taken together, these restrictions ensure we retain well-attached workers for whom we can credibly infer the human capital accumulation parameters from the model. Over the period 1975–2014, these conditions leave us with around 150 thousand workers employed at around 250 thousand firms.

3.2 Model implementation

We set the unit of time t to a month. We assume that output per period, y_t, in a match between a firm with fixed productivity θ and a worker who has accumulated specific and general human capital, s_t and g_t, and with current match productivity ε_t, is given by

$$y(\theta, g_t, s_t, \varepsilon_t) = \theta \cdot g_t \cdot s_t \cdot \varepsilon_t. \tag{12}$$

We further assume that the sampling distribution of level productivity θ is log-normal with mean zero and standard deviation σ_θ and that the idiosyncratic component of match productivity ε follows an AR(1) process in logs

$$\ln \varepsilon_t = \rho_\varepsilon \ln \varepsilon_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \sigma_\varepsilon). \tag{13}$$

The initial productivity ε_0 in newly formed matches is set to the median value of the unconditional distribution of ε.[16] The grid for general human capital, g, is uniformly spaced in logs with seven points on the interval $[0, \ln g_{n_g}]$. Similarly, the grid for specific human capital s has seven equidistant (in logs) points in the interval $[0, \ln s_{n_s}]$.[17] Finally, we assume that the flow utility of being nonemployed is proportional to the level of general skills accumulated by the worker $z(g_t) = b \cdot g_t$.

[15]In the SIAB data, the schooling variable is frequently missing or misreported. We rely on the imputation procedure described in Fitzenberger, Osikominu and Völter (2005) to improve the quality of the education measure.

[16]Or, similarly, $\varepsilon_0 = \exp(\mathbb{E}[\ln \varepsilon_t])$.

[17]Together, the restrictions on θ, g, s, and ε normalize the ex-ante log productivity of a draw for new labor market entrants to zero. This normalization is required because the average productivity level is not pinned down in the estimation.

3.3 Estimation strategy

The model generates transitions both in and out of employment and between employers, as well as rich wage dynamics. It is estimated by targeting a combination of moments from the data and estimates from reduced-form regressions. In total, we target twenty-one moments to estimate fourteen parameters. Although all model parameters are estimated jointly, we heuristically link each parameter to one of its most informative moments in the description of our estimation strategy. Appendix D provides additional results on the link between the model parameters and moment targets.

Transition parameters In line with the model and the analysis of losses detailed in Section 4 below, we do not make a distinction between unemployment and inactivity. We simply treat all gaps between employment spells as nonemployment spells and define the corresponding transition rates accordingly. A detailed description of the construction of all variables used in the quantitative section is provided in Appendix A.2.

The parameters λ_E and λ_N governing, respectively, job-to-job transitions and those from nonemployment to employment are closely related to the EE and NE transition rates in the data. An increase in the contact rate during employment increases the probability of job switching, and a higher contact rate during nonemployment increases the probability of NE transitions.

The observed rate of separation into nonemployment EN helps us discipline the rate δ at which matches are hit by an exogenous job destruction shock. In the model, high-tenure workers, who have both high-θ and high-s jobs, are unlikely to be separated endogenously in response to idiosyncratic shocks ε. Because there are high-tenure workers making EN transitions in the data, the exogenous separation rate δ can be pinned down separately from the parameters governing the idiosyncratic shock process (13).

The parameter λ_R, which governs the rate at which workers hit by a δ-shock sample a new job offer from $F(.)$ without transiting through nonemployment, is informed by the average change in the wage rate following a job-to-job transition. This statistic is informative about relocation shocks in our setting because the accumulation of specific skills, which are forgone in the case where workers switch to a different employer, raises the productivity threshold required for an EE transition to take place. Setting $\lambda_R = 0$, the model would predict a much larger wage increase following an EE transition than what is found in the data.

The parameter κ, which governs the exit rate from the labor market, is set to match the average potential experience observed in the data. We set κ to approximate a mean potential experience of 16.5 years.[18]

[18]Because the data only cover private sector employees, attrition can have several different origins in our

Workers' bargaining power To inform the bargaining power parameter α, we follow Jarosch (2023) and use information on the difference between the average log-wage of hires from nonemployment and the average log-wages of all employed workers.[19] This statistic is informative about workers' bargaining power in our model because the initial wage of hires from nonemployment w_0 is determined by Nash bargaining

$$w_0 : \; W(\theta, s_0, g, \varepsilon_0, w_0) = U(g) + \alpha S_0(\theta, g).$$

As α gets larger, the disadvantage of newly hired workers diminishes, implying that the difference between the wages of new and existing workers shrinks.

Idiosyncratic component of match productivity In the model, more productive matches last longer and are more likely to survive negative idiosyncratic ε-shocks. This feature implies that the model generates declining probabilities of separation into nonemployment by tenure. We therefore use the yearly tenure profile of separations EN to identify the parameters governing the distribution of the idiosyncratic component of match productivity $H(\varepsilon'|\varepsilon)$.[20]

Sampling distribution of firm productivity Wage dispersion helps identify the parameter controlling the variance of the sampling distribution of the fixed component of employer productivity σ_θ. To inform this parameter, we target the mean-min wage ratio on residualized log-wage data, where residual wagers are obtained from a regression on year and worker effects.[21] Firm productivity θ plays a key role in determining wages in the model, along with workers' human capital.

General and specific human capital The parameters related to general and specific human capital are disciplined using wage moments. Matched employer-employee data are key in this case, since they allow us to separately identify the role of specific and general human capital from the job ladder as wage determinants. Employer identifiers are therefore needed to retrieve firm effects.

Reduced-form estimates of returns to experience and tenure from a model that controls for firm fixed effects allow us to retrieve information on the accumulation rate of each type of

sample, such as retiring, taking a job in the public sector, or becoming self-employed.

[19] We first take out year effects and worker effects from log wages in the data because these drivers of wages are absent from our model.

[20] Krolikowski (2017) uses a similar identification strategy in a model with no skill accumulation.

[21] We follow Hornstein, Krusell and Violante (2011) and exponentiate the wage residuals and compute the ratio of the mean to the fifth percentile. In our setting, we only control for year and worker effects to purge the data of two key forces driving wages that are absent from our model.

skills. The inclusion of a firm fixed effect in this regression model controls for the role of the job ladder. The returns to tenure and experience derived from a two-sided fixed effects model are used to inform the parameters governing the accumulation of general human capital during employment ($\ln g_{n_g}$ and ϕ_E) and specific human capital during workers' tenure at the same firm ($\ln s_{n_s}$ and γ). Specifically, we estimate the following Mincer equation

$$\ln w_{it} = \sum_{k=1}^{2} \xi_k \cdot \mathrm{EXP}_{it}^k + \sum_{k=1}^{2} \zeta_k \cdot \mathrm{TEN}_{it}^k + \alpha_i + \psi_{j(i,t)} + \epsilon_{it}, \tag{14}$$

where the log-wage of individual i in month t is regressed on a quadratic polynomial of actual experience (EXP) and tenure at the current employer (TEN), an individual fixed effect, and a firm fixed effect $\psi_{j(i,t)}$. ϵ_{it} is the residual. We then use the estimated coefficient $\{\hat{\xi}_k, \hat{\zeta}_k\}_{k=1}^2$ as additional moment targets. In practice, we cluster the firm fixed effects $\psi_{j(i,t)}$ for two reasons. First, the limited mobility of workers between employers could make the estimated returns to tenure and experience with standard firm fixed-effects imprecise. Second, in the SIAB-7514 dataset, we observe only 2 percent of the total population of German workers. Therefore, using the regular employer identifier would imperfectly control for firms' time-invariant characteristics. Following Bonhomme, Lamadon and Manresa (2019), we use a k-means algorithm to group employers in a first step, based on the average wages they pay to their workers, and use the obtained group identifiers as a proxy to compute the corresponding employer fixed effects in a second step.[22] Appendix D reports additional results showing that general and specific human capital are required to match the experience and tenure profiles derived from Equation (14).

We also include the EE transition profile by tenure as an additional source of identification for the accumulation of specific skills. The EE-tenure profile is closely linked to the accumulation of specific skills, since the latter implies that the incentive to switch jobs declines with tenure at the firm. Conditional on the sampling distribution of firm productivity, the steeper (flatter) the EE-tenure gradient, the faster (slower) is the accumulation rate of specific human capital.

Finally, to inform the parameter that governs the rate of decay of general human capital during nonemployment (ϕ_N), we estimate the regression

$$\ln w_{it}^0 = \pi \cdot \mathrm{DUR}_{it} + \alpha_i + d_t + \epsilon_{it}, \tag{15}$$

where w_{it}^0 denotes the first wage recorded after a nonemployment spell, DUR_{it} is the length of the nonemployment spell in month, and α_i and d_t are individual and year fixed effects.

[22] See Appendix A.4 for implementation details.

The estimated coefficient $\hat{\pi}$ is used as an additional moment target.

The model moments are derived from a simulated panel of worker histories similar to the actual data. In computing the moments from the simulated panel, we closely replicate the steps to obtain the moments computed on the actual data. Details on the solution and simulation of the model can be found in Appendix B.

3.4 Model fit

We report the value of the moments derived from the SIAB data, along with their model-generated counterpart, in Table 1 and Figure 1. The estimated parameters are shown in Table 2.

Table 1: Fit to targeted moments

	Model	Actual
NE monthly transition rate	0.087	0.052
EE monthly transition rate	0.007	0.010
EN monthly transition rate	0.007	0.008
Separation rate by tenure		
EE separations	See Figure 1b.	
EN separations	See Figure 1a.	
Returns to tenure and experience (eq. 14)		
Tenure polynomial $\{\hat{\zeta}_k\}$	See Figure 1c.	
Experience polynomial $\{\hat{\xi}_k\}$	See Figure 1d.	
Mean-Min wage ratio	1.277	1.280
$E(\ln w \vert \text{NE}=1) - E(\ln w)$	-0.138	-0.109
Non-employment duration coef. (π in eq. 15)	-0.002	-0.002
$E(\Delta \ln w \vert \text{EE}=1)$	0.054	0.048

Notes: The table shows the targeted moments. All moments are derived from the monthly panel data described in Section 3.1 and Appendix A. The transition probabilities are monthly rates. The separation by tenure profiles and the wage profiles by experience and tenure are reported in Figure 1. All wage statistics are purged of year and worker effects in the data. See main text for details on the construction of these moments.

Overall, the model fits well the moments shown in Table 1. It is broadly in line the average rates at which workers find jobs, both from nonemployment (NE) and employment (EE), as well as the rate at which workers lose jobs (EN). It also accurately replicates the negative relationship between the job separation rates (both EN and EE) and tenure (Figures

Table 2: Estimated model parameters

Parameter	Description	Value
σ_θ	Firm type: $\theta \sim \ln \mathcal{N}(0, \sigma_\theta)$	0.057
ρ_ε	Persistence AR(1) for match-specific shocks (13)	0.843
σ_ε	Standard dev. AR(1) for match-specific shocks (13)	0.082
λ_N	Contact rate non-employment	0.189
λ_E	Contact rate employment	0.089
δ	Exogenous job destruction rate	0.003
$\ln s_{n_s}$	Max level of firm-specific human capital	0.218
$\ln g_{n_g}$	Max level of general human capital	0.362
γ	Appreciation rate firm-specific human capital	0.014
ϕ_E	Appreciation rate general human capital	0.030
ϕ_N	Depreciation rate general human capital	0.044
α	Worker bargaining weight	0.829
b	Home production factor: $z(g) = b \cdot g$	1.207
λ_r	Reallocation shock rate (conditional on δ-shock)	0.345

Notes: The table shows the estimated model parameters. The parameters are estimated jointly using the simulated method of moments. See Appendix B.2 for details.

1a and 1b). For instance, the monthly EN transition rate drops by 2 percentage points in the first three years of tenure, while the monthly EE transition rate is reduced by one percentage point over the same time frame.

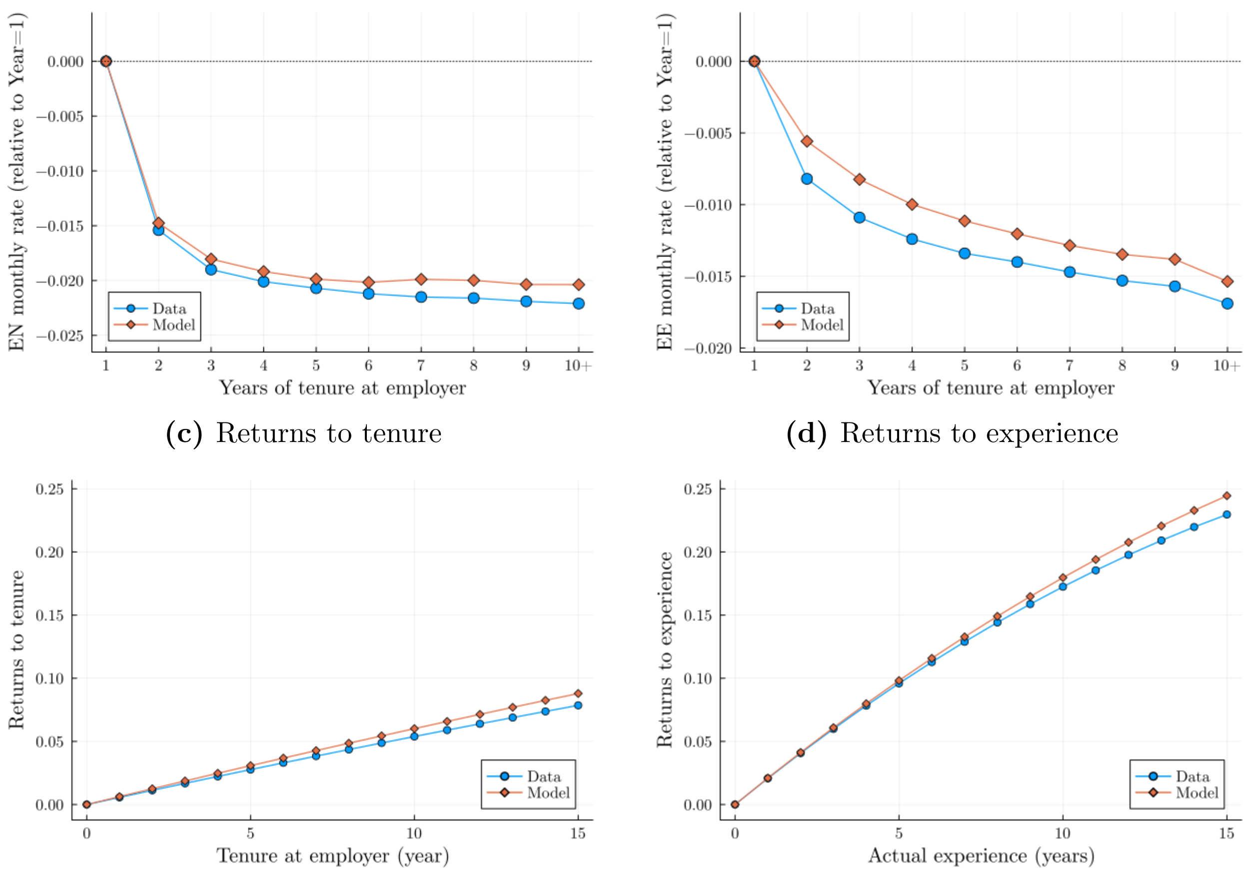
Figure 1: Model fit of targeted moments

(a) Separation to nonemployment (EN)

(b) Separation to employment (EE)

(c) Returns to tenure

(d) Returns to experience



Notes: The figure shows the model fit to key targeted statistics. Panels (a) and (b) show the average monthly separation rate to nonemployment (the EN rate) and to another employer (the EE rate) by tenure in years relative to the first year, so the y-axis shows the percentage difference in the separation rate relative to the first year of tenure. Panels (c) and (d) show the implied returns to tenure and to experience estimated from Equation (14) on actual and model-simulated data.

The model generates the correct amount of wage dispersion, as measured by the mean-min wage ratio. The calibrated value for the standard deviation of the fixed component of the firm productivity distribution, $F(\theta)$, is equal to 0.057. This is much lower than the 0.37 estimate in Krolikowski (2017), because, in our model, general and specific skills contribute to wage dispersion and wage growth in addition to firm productivity. It also reproduces the negative relationship between entry wages and time spent in nonemployment estimated in the data. In the data one more month spent in nonemployment is associated with a reduction in (log) wages equal to 0.2%, which the model matches exactly. Finally, the model delivers a very close fit of the returns to tenure (panel c) and experience (panel d) estimated using the data. The corresponding parameters translate into structural yearly accumulation rates of general and specific human capital equal to 1.5% and 0.6%, respectively, and a depreciation rate of general human capital equal to 0.8% per year.[23] These structural returns are different from the empirical estimates reported in the empirical literature focused on obtaining unbiased returns to experience and tenure for realized wage growth, accounting for selection (Altonji and Shakotko, 1987; Topel, 1991). In a bargaining model like the one developed here, workers do not capture the full productivity gains from the match by construction, so realized wage growth likely underestimates the true returns.

Appendix E shows that the model can reproduce several additional statistics not directly targeted in the calibration. Notably, the model is broadly consistent with the EN and EE separation by tenure profiles during workers' first 24 months with a new employer. It also replicates closely the large mass of job stayers with no or little monthly wage growth, which we interpret as a validation of the sequential auction wage-setting protocol detailed in Section 2.2.[24] There are also dimensions of the data along which the model performs less well. For instance, the model tends to underpredict longer nonemployment spells, with 95% of workers re-employed within three years against 80% in the data.

3.5 The role of specific human capital

specific human capital sets this paper apart from prior work studying the contribution of the job ladder and general human capital to post-displacement earnings losses (Jung and Kuhn, 2019; Burdett et al., 2020; Jarosch, 2023). Among those, only Jung and Kuhn (2019) allow for some degree of specific human capital accumulation by assuming that a job transition implies an expected loss of general human capital independently of job tenure. This form of irreversibility is effectively a tenure-independent switching cost and has different implications than the standard, tenure-dependent, specific human capital accumulation considered in this paper (Becker, 1964; Topel, 1990).

There is one testable implication that distinguishes our framework from these three papers. In the absence of specific human capital accumulation, the only sources of correlation between job tenure and workers' EN and EE transition rates are: (i) selection based on workers' or

[23]The structural returns are defined as $12 \cdot \Delta \ln x \cdot \omega_x$ where $\Delta \ln x$ denotes the step size of the (equally-spaced) human capital grid and ω_x the fraction of simulated observations not at the grid boundary. The ω_x term accounts for observations with zero return to human capital by construction, such as those with g_0 in nonemployment.

[24]We show in Appendix F that using Nash-bargaining instead of sequential auction implies a much larger wage growth dispersion for stayers because the match-specific shock ε is passed directly into wages.

employers' characteristics, or (ii) correlation between labor market experience and job tenure. By contrast, specific human capital accumulation implies a negative relationship between tenure and separation rates even controlling for worker and employer characteristics and for experience. This is because, all else equal, longer job tenure is associated with higher productivity in the current job, thus forming the basis for an empirical test of this implication of our model.

To isolate the role of specific human capital accumulation in our setting, we consider a series of restricted versions of the model. Starting from a model with a single match-specific shock (ε only), we incrementally switch on permanent firm heterogeneity (ε and θ) and general human capital (ε, θ and g).[25] We then compare the conditional separation by tenure profiles generated by each version of the model to the one obtained from the data. Specifically, we estimate the regression models

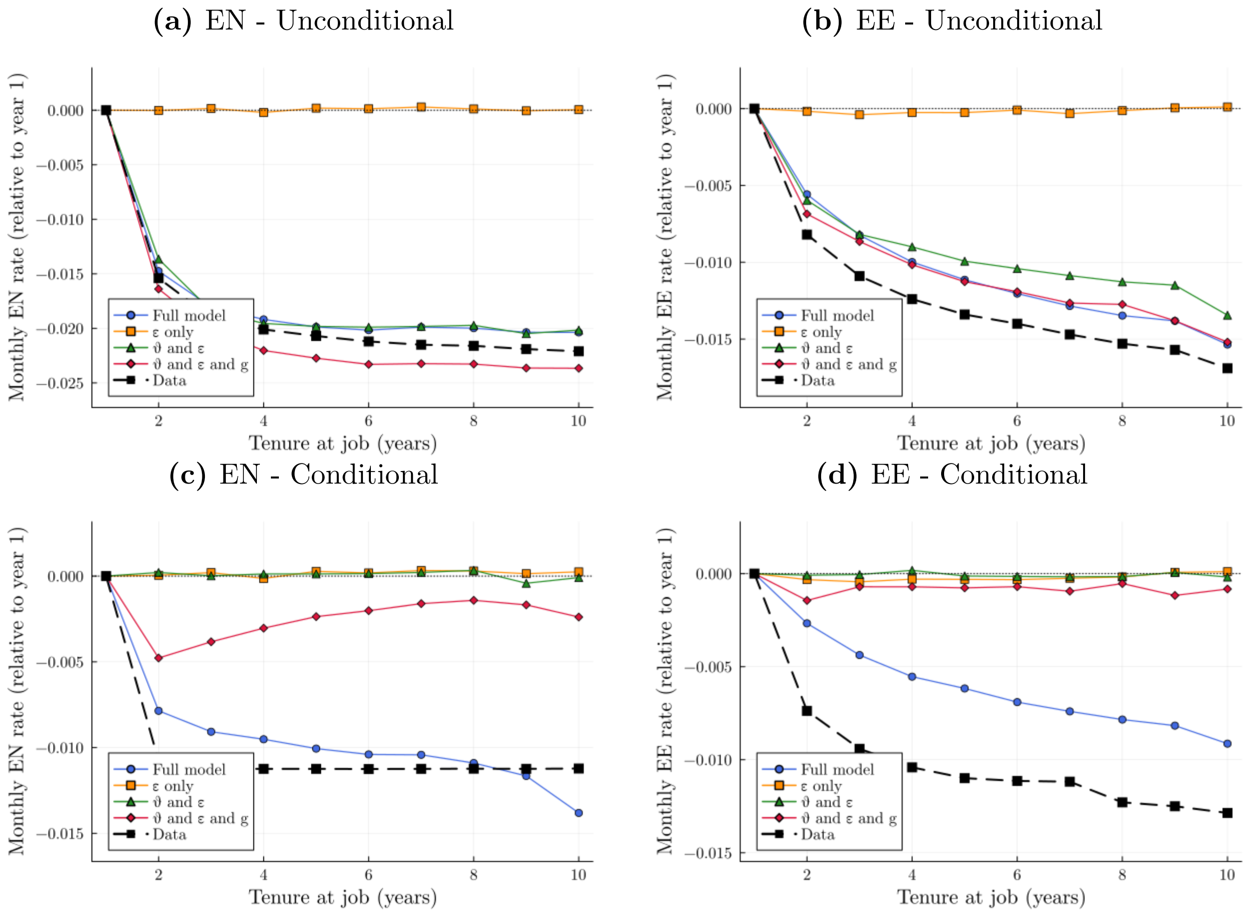
$$\mathrm{SEP}_{it} = \text{TEN-YR}_{it} + \sum_{k=1}^{2} \xi_k^{\mathrm{SEP}} \cdot \mathrm{EXP}_{it}^k + \alpha_i + \psi_{j(i,t)} + \epsilon_{it}, \tag{16}$$

where SEP_{it} is an indicator for a separation of type EN or EE, TEN-YR_{it} is a vector of dummies for years of tenure at the current employer, EXP_{it} is potential experience, and $\psi_{j(i,t)}$ are clustered firm fixed effects. The estimated coefficients on the tenure-year dummies in Equation (16) represent our measure of conditional separation rates by tenure. These profiles are the counterparts of the unconditional separation by tenure profiles reported in figures 1a and 1b. We estimate Equation (16) separately for EN and EE transitions, and on both the actual data and each version of the model-simulated data. For comparison, we also report the corresponding unconditional EN and EE separation by tenure profiles.

The results are shown in Figure 2. Except for the model with a single match-specific shock ε, all models can match the unconditional EN and EE tenure profiles derived from the data, which are targeted in the estimation (figures 2a and 2b). The ε-only result is not surprising because all jobs are the same in expectation in that model by construction, so the survival threshold is not correlated with tenure. Two results stand out once we add controls for experience and for worker and firm heterogeneity in Equation (16). First, the data still show a negative separation profile by tenure for both EN and EE transitions, indicating that the controls do not fully account for the unconditional EN and EE profiles. Second, only the full model with specific human capital clearly generates these profiles, which are not targeted in the estimation (figures 2c and 2d). The model with firm heterogeneity and general human capital but no specific human capital accumulation (ε, θ and g) also gives rise to a negative

[25] All model parameters are re-estimated for each version of the model, targeting the same moments as in the baseline model. See Appendix C for details.

Figure 2: Unconditional and conditional effect of job tenure on separation rates



Notes: The figure shows the derived separation by tenure profiles for EN and EE transitions in the data and in the model-simulated data for different versions of the model. The unconditional profiles are obtained from a regression of the separation indicators on tenure-year dummies (panels a and b). The conditional profiles are obtained from a regression of the separation indicators on tenure-year dummies in Equation (16), with the controls described in the main text (panels c and d). The profiles are normalized to zero in the first year of tenure, so the y-axis shows the percentage-point difference in the separation rate relative to the first year of tenure.

relationship between tenure and EN separation rates, but it is much less pronounced than in the data. Similarly, only the full model with specific human capital accumulation can replicate the negative relationship between tenure and EE separation rates from the data once we control for experience and for firm and worker heterogeneity.

An important caveat in interpreting the results in Figure 2 is that Equation (16) is only an approximation of the functional form underlying endogenous separations in each version of the model since (i) some of the state variables are either omitted (e.g., ε) or imperfectly measured (e.g., experience as a proxy for g), and (ii) the corresponding relationships are not necessarily linear.[26] This explains why the conditional EN separation by tenure profile generated by the model with firm heterogeneity and general human capital (ε, θ and g) is not necessarily flat despite the absence of specific human capital accumulation in that model. Yet, within our framework, the results in Figure 2 strongly suggest that specific human capital accumulation is required to account for the conditional separation by tenure profiles found in the data.

More generally, the literature has emphasized learning about match quality as an alternative explanation for the separation and wage-tenure profiles observed in the data (Jovanovic, 1979). Learning about match quality and specific human capital accumulation are difficult to distinguish using worker-level data alone since both imply (i) a negative relationship between tenure and separations, and (ii) a positive relationship between tenure and wages. One way to disentangle these two mechanisms is to obtain data on individual workers' output. For instance, Pastorino (2024) uses data on managers at a single firm to quantify these forces within a structural model, but the data requirements of this approach make it difficult to generalize. Alternatively, the seminal contribution of Nagypál (2007) disentangles specific human capital accumulation from learning about match quality using level shocks, which are shown to have different implications for the separation-tenure profile under each mechanism. This identification strategy cannot be applied directly in our environment because (i) it requires level data, and (ii) it is restricted to ex ante identical firms.[27] Extending this approach to richer environments such as the one developed here is an interesting avenue for future work.

[26] Concretely, in the (θ, ε, g)-model, the conditional regression is an imperfect proxy for $S(\theta, \varepsilon, g) < U(g)$.

[27] While (i) is a data requirement, (ii) poses a challenge in an environment with firm heterogeneity because constructing a model counterpart to level shocks requires specifying firm boundaries. Nagypál (2007) circumvents this issue by simulating workers at a single firm over a long period and treating the resulting time series as a cross-section. To the best of our knowledge, search models with large firms maintain the assumption that all workers are identical (Elsby and Gottfries, 2022; Audoly, 2026).

4 The cost of job loss

4.1 Reduced-form analysis

We construct a displacement sample following the standard approach in the literature (e.g., Davis and von Wachter, 2011). We aggregate our data at the yearly level and select a sample of high-tenured workers (workers with more than three years of tenure) in the yearly panel.[28] In each separation year Y, we only consider prime-age workers (defined as workers with 5 to 34 years of potential experience) who, in addition, are continuously employed in years $Y-1$, $Y-2$, and $Y-3$ with the firm recorded in Y.[29] The treatment group is made up of workers who experience a separation into nonemployment from their long-term employer in year Y and who are re-employed in a different firm by year $Y+3$. The control group is made up of workers who did not experience a separation from their long-term employer in year Y.

We follow the standard approach in the literature to estimate workers' losses from displacement.[30] We estimate the event-study regression

$$y_{it}^{Y} = \alpha_i^{Y} + d_t^{Y} + \beta X_{it}^{Y} + \sum_{k=-5}^{10} \delta_k \cdot D_{it}^{Y,k} + \epsilon_{it}^{Y}. \tag{17}$$

The variable y_{it}^{Y} denotes the outcome of interest (log-earnings and log-wages) for individual i in calendar year t for displacement year Y. The worker effect α_i^{Y} captures worker heterogeneity, d_t^{Y} represents a year fixed effect, and the vector X_{it} is a cubic polynomial in potential experience for individual i at time t. The indicator $D_{it}^{Y,k} = \mathbf{1}[t-Y=k,\ \mathrm{EN}_{i,Y}=1]$ equals one if calendar year t is k periods relative to displacement year Y and worker i was displaced in year Y. We use the convention that $k=0$ denotes the separation year, so $k=0$ is the last year of positive earnings with the pre-displacement employer, and $k=1$ is the first year with zero earnings from the pre-displacement employer. For example, when estimating earnings losses for displacement year $Y=1985$, $D_{i,1985}^{Y,0}$ is equal to one in year $t=1985$ if worker i experiences displacement during that year, and equal to 0 in all other years $t \neq Y$. $D_{i' \neq i,t}^{Y,k}$ is equal to zero in all t for all other individuals who belong to the sample and did not experience displacement in year Y.

We follow Flaaen et al. (2019) and Jarosch (2023) and estimate equation (17) by stacking all possible displacement years between 1981 and 2005 to obtain the coefficients $\{\hat{\delta}_k\}$. These coefficients therefore measure the evolution of the variable of interest before and after

[28]See Appendix A.3 for details.

[29]We use potential experience instead of age in our definition to be consistent with the model.

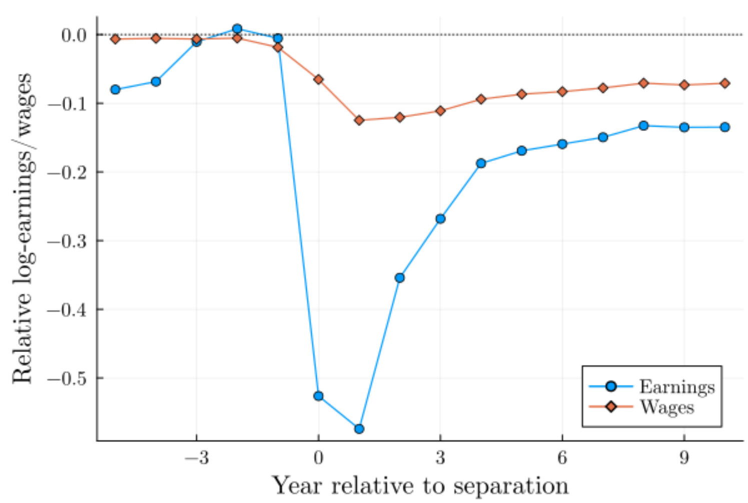
[30]See, for example, Davis and von Wachter (2011), Flaaen et al. (2019), Lachowska et al. (2020) and Schmieder et al. (2023).

separation in year Y relative to the baseline year $k = -6$ and relative to the control group. This estimation strategy treats all potential separation years as separate data sets, as reflected in the notation. For example, the worker effect α_i^Y is specific to a worker i and a separation year Y.

An alternative approach put forward in the literature is to run specification (17) year-by-year for each separation year Y and average across separation years to obtain the corresponding losses (see, for instance, Davis and von Wachter, 2011). We choose the "stacked" empirical strategy for two reasons. First, given our sample selection criteria and our data, the number of separations in any given year is limited. Second, as noted by Flaaen et al. (2019) and Jarosch (2023), this approach directly yields standard errors for the estimated coefficients $\{\hat{\delta}_k\}$ specified in (17). We again follow their methodology and two-way cluster standard errors at the person and year level.

We plot the coefficients for wages and earnings estimated on the SIAB-7514 data in Figure 3. The results are in line with those found for Germany in prior work (Burdett et al., 2020; Schmieder et al., 2023; Jarosch, 2023). Wages drop by more than 10 log-points and recover only very gradually; they are still 6-7 log-points lower than in the control group ten years after the separation event occurs. Earnings exhibit a very large drop upon separation, followed by an initially swift recovery that becomes much slower three to four years after the separation event,mirroring the pattern for wages. The persistence of earnings losses is therefore largely driven by the persistence of wage losses.

Figure 3: Post-displacement earnings and wage losses in the data



Notes: The figure shows the estimated event-study coefficients $\{\hat{\delta}_k\}$ from Equation (17). The "Earnings" specification uses yearly log-earnings as the dependent variable. The "Wage" specification uses log-wages as the dependent variable.

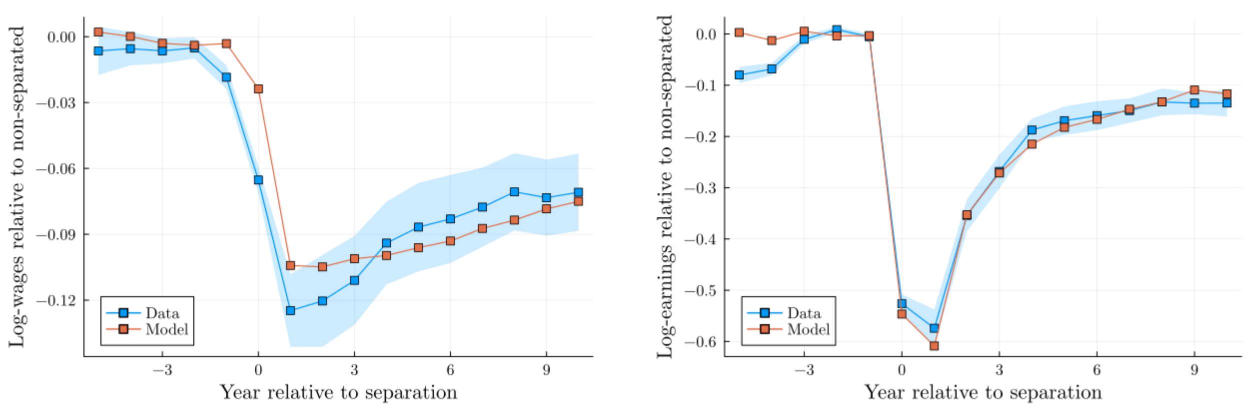
4.2 Model versus data

We compare the earnings and wage losses in the data with their counterparts in the model simulation. The simulated losses are estimated by applying the same sample selection and estimation method as for the empirical ones. The key difference is that individual fixed effects and year fixed effects are omitted since the model is stationary and does not feature individual heterogeneity.

The results of this comparison are shown in Figure 4. Overall, the model replicates the drop and recovery in wages and earnings very well. The persistence of wage losses in the model is very similar to that of their data counterpart. A small discrepancy between the wage losses generated by the model and those measured in the data can be noted prior to displacement. Wages start to drop before displacement in the data, most likely due to wage freezes or reductions associated with the separation to come. While this mechanism is potentially present in the model, since match-specific shocks (ε_t-shocks) can trigger a downward wage renegotiation, it does not exhibit the same magnitude as in the data.

Figure 4: Fit of losses

(a) Wage losses **(b)** Earnings losses



Notes: The figure shows the estimated $\{\hat{\delta}_k\}$ in Equation (17) on actual and model-simulated log wage data (panel a) and log-earnings data (panel b). The blue shading denotes the 95 percent interval two-way clustering standard errors at the person and year level.

4.3 Structural decomposition of wage losses

In the model, job search, general human capital, and specific human capital are the three key forces that can jointly explain the loss in wages of separated workers. To quantify the relative contribution of each of these forces, we use the model to build counterfactual wage series for workers who experience a separation event. Following the approach described in

Jarosch (2023), we do so in a stepwise fashion.

Step 1: We define the treatment group as all high-tenure workers who are exogenously separated (due to a δ shock) in year Y of our simulation. We build a control group by artificially preventing these separations (by setting $\delta = 0$ for the treated workers in year Y) and repeat our simulation procedure using otherwise identical shocks.[31] This simulation generates a counterfactual series for wages, employment, general skills g, specific skills s, as well as employer productivity θ in the years $\{Y, Y+1, \ldots, Y+10\}$ following separation. By construction, the series are the same for the treated and counterfactual groups in the years prior to separation.

Step 2: We let treated workers artificially retain general human capital. To be specific, upon re-employment we assign the general human capital (g) they would have had if they had not been separated. We repeat our simulation procedure for treated workers, but now with the g from the control group. The difference between the wage losses of the treated workers under Step 1 and those of this counterfactual group is a measure of the contribution of g to overall wage losses.

Step 3: We use the exact same procedure as in Step 2, but now upon re-employment we assign both the general *and* specific human capital that workers in the control group have. The difference between the wage losses in Step 2 and those in this counterfactual with both g and s set to the control group's values is a measure of the contribution of specific skills s to the overall losses.

Step 4: We use the exact same procedure as in Steps 2 and 3, but now we assign the general human capital, specific human capital, and firm productivity that workers in the control group have upon re-employment. The difference between the wage losses in Step 3 and those in this counterfactual with g, s, and θ set to the control group's values is a measure of the contribution of θ to the overall losses.

By construction, the counterfactual workers in Step 4 have, on average, the same surplus as the counterfactual workers in the control group. Recall from Equation (11) that the surplus does not depend on how the wage splits the match output between workers and firms. However, wages may still differ between the counterfactual workers in Step 4 and the control group. The reason is that workers in the control group have potentially accumulated additional bargaining rents by using outside offers to renegotiate their wages. The residual difference is, therefore, a measure of the contribution of these rents to the overall losses.

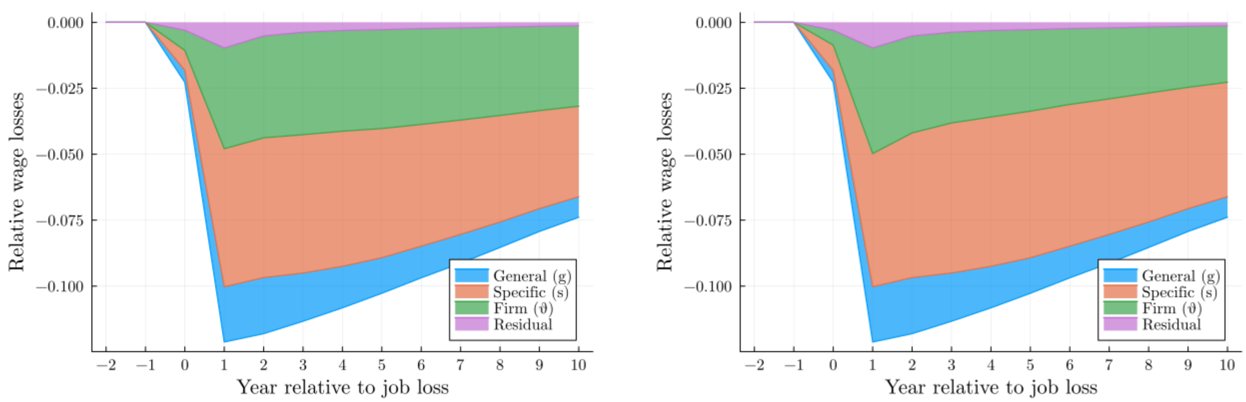
We compute the contribution to wage losses of the three channels in Steps 2 to 4 by

[31]We perform the counterfactual for exogenously separated workers because it is unclear how to prevent separations in a consistent way for endogenously separated workers given the persistence of match-specific shocks. In practice, most separations in the simulated displacement data are exogenous.

regressing the simulated log-wages series in the treated group and in each counterfactual group using the same event-study specification as in Equation (17), where the control group is now defined as in Step 1.

Figure 5: Structural decomposition of wage losses

(a) Baseline **(b)** Alternative



Notes: The figure shows the structural decomposition of wage losses obtained following the steps outlined in Section 4.3. Baseline counterfactual wage losses are obtained by sequentially removing the loss of general human capital, then specific human capital, and finally firm productivity (panel a). Alternative counterfactual wage losses are obtained by sequentially removing the loss of firm productivity, then specific human capital, and finally general human capital (panel b).

The wage function implied by the model at the estimated parameters is not log-linear. As a result, the order in which we construct the counterfactual series affects the contribution of each component to the overall losses.[32] Figure 5 reports two alternative implementations of the structural decomposition implied by the model. In panel (a), we build counterfactual wage losses by first assigning the control group's specific human capital s and then firm type θ. We do the opposite in panel (b).

A robust finding that emerges from this exercise is that the loss of a worker's specific human capital is the most important source of wage loss, especially in the short term (46–52% of cumulated losses). The loss of specific capital is the second key factor behind the size and persistence of wage losses (30–36% of cumulated losses). Both general human capital (15% of cumulated losses) and bargaining rents (less than 3%) are second-order factors.[33] Through the lens of the model, the two components specific to the employer and directly entering the

[32] In Appendix G, we present a robustness exercise in which we experiment with various permutations of the state variables (g, s, θ) that can be used to build the counterfactuals described above. We find that our main conclusions are robust to alternative permutations.

[33] This last finding on bargaining rents is similar to the results in Jarosch (2023). While this quantitative result makes the sequential bargaining auction protocol less central to the decomposition, this channel could matter in other calibrations of the model because of the time required for workers to capture bargaining rents.

production technology, s and θ, therefore account for most of the size and persistence of wage losses.

5 Conclusion

The main contribution of this paper is to provide a quantitative framework that can account for the relative strength of the forces driving the cost of job loss. To do so, we build a model in which wage gains come from three sources over the course of workers' career: (i) searching for a better employer, (ii) accumulating specific skills, and (iii) accumulating general skills.

We use matched employer-employee data from Germany to compute moments related to job mobility and wage growth to discipline the process of job search and the accumulation rates of general and specific skills. The estimated model can replicate the long-term losses in earnings and wages experienced by displaced workers.

A series of counterfactual experiments suggest that the loss of specific human capital and the loss of the employer premium are the main drivers of wage losses. About half of the cumulative wage losses experienced by displaced workers are due to the loss of specific skills and about one-third to the loss of a job with a good employer. The lower rate of general human capital accumulation accounts for nearly all the remainder.

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Online Appendix

A Data construction

A.1 Construction of the monthly panel

The SIAB data set contains information about the employment history of every individual in the sample stored in spell format with given start and end dates. We transform the data set from spell format to monthly format. We do this by choosing the 1st of the month as the reference date and attributing the information of the spell to the month if the spell starts before or on the 1st of the month. For example, if the worker is employed full-time subject to social security in the spell that goes from the 29th of January until the 15th of March, we assign this information to the months of February and March. The monthly panel consists of approximately 31 million observations.

A.2 Variables definition

Employment and nonemployment Workers are defined as employed in month t if they are employed full-time subject to social security on the first day of the month. Workers are considered nonemployed in all other cases, including if they work part-time. In months when workers do not have a record in the SIAB, they are only considered nonemployed in-between two spells.

Wages and earnings Wages are recorded only for employed workers, and are considered missing for nonemployed workers. Earnings are equal to wages during months of employment and to zero during months of nonemployment.

Job-to-job transition (EE) A job-to-job transition is recorded in the following two cases: (i) if the worker is employed in firm j in month t and in firm j' in month $t+1$; (ii) if the worker is employed in firm j in month t and in firm j' in month $t+2$, and the worker is nonemployed and does not apply for unemployment benefits in month $t+1$. The second part of the definition allows for a gap longer than one month in-between two jobs on the condition that the worker does not apply for nonemployment benefits.

Nonemployment to employment transition (NE) A nonemployment to employment transition is recorded in the following two cases: (i) when the worker is nonemployed in month t for at least two periods and employed in month $t+1$; (ii) if the worker is nonemployed in month t and applies for unemployment benefits and employed in period $t+1$. The second part of the definition is a direct implication of the definition of job-to-job transitions, which allows for a one-month gap between two jobs on the condition that workers are not claiming unemployment benefits.

Employment to nonemployment transition (EN) An employment to nonemployment transition is recorded in the following two cases: (i) when the worker is employed in month t and nonemployed for exactly one month and applying for unemployment benefits in month $t+1$, and (ii) if the worker is employed in month t and nonemployed for at least two periods. The first part of the definition is a direct implication of the definition of job-to-job transitions, which allows for a one-month gap between two jobs on the condition that workers are not claiming unemployment benefits.

A.3 Construction of the yearly panel

Starting from the monthly dataset, we transform the employment, earnings, and wages variables into yearly observations by averaging the records across all months during a year. We record an employment-nonemployment transition (EN) and a job-to-job transition (EE) in a given year, respectively, if at least one EN or EE transition is observed in the monthly panel in that year. We consider the annual employer the establishment in which the worker is employed in January of the corresponding year. The yearly panel consists of approximately 2 million observations.

A.4 Unobserved firm heterogeneity

To account for firm heterogeneity, we follow the recent literature based on the work by Bonhomme et al. (2019) and group firms into 20 groups using a k-means algorithm. We cluster firms based on their wage distribution and use the group identifiers as controls in the two-sided wage regression (Equation 14). The idea is that variation in the wage distribution at the employer level conveys information about the employer's underlying unobserved "type." In practice, we implement the classification based on the average log wages paid by firms to full-time workers (in line with our sample selection criteria).[34]

[34]We use the residual of a regression of firms' average log wages on year dummies to net out the time variation.

B Model solution and estimation

B.1 Model solution

We solve the model numerically under the assumptions listed in Section 3.2. In practice, we jointly solve the value functions for the worker surplus (11) and the nonemployed worker (8) on a discretized grid for the state variables $(\theta, s, g, \varepsilon)$.

There is no closed form for the wage function, and we derive it numerically.[35] We build a wage grid and solve for the value function for employment (9) by value function iteration, conditional on the value functions for the worker surplus and nonemployment. The wage function is then obtained by inverting this function using the bisection method in accordance with the bargaining protocol rules described in Section 2.2.

We then simulate data from the model at monthly frequency. Specifically, we simulate work histories for 15,000 workers, all starting nonemployed, for eighty years. We discard the first forty years to remove the effects of initial conditions and compute the moments needed for identification on the remaining forty years. In the simulation, we allow the time-varying idiosyncratic shock component ε to take values in between grid points, but not above and below the minimum and maximum values on the grid.

B.2 Estimation

We use the simulated method of moments to obtain the model parameters. As explained in Section 3.3, we compute the same set of moments on the actual and model-simulated data. The vector of estimated model parameters $\hat{\Xi}$ solves

$$\hat{\Xi} = \underset{\Xi}{\operatorname{argmin}} \big[\hat{m} - \tilde{m}(\Xi)\big]^T \hat{\Omega} \big[\hat{m} - \tilde{m}(\Xi)\big], \tag{18}$$

where $\hat{m}$ represents the vector of N_m data moments, $\tilde{m}(\Xi)$ represents the vector of N_m model-simulated moments, $\hat{\Omega}$ is an $N_m \times N_m$ weighting matrix, and Ξ denotes the vector of N_Ξ parameters.

The weighting matrix $\hat{\Omega}$ is diagonal with typical element $[\hat{\Omega}]_{jj} = \omega_j / \hat{m}_j^2$. We use subjective weights $\omega_j > 0$ to closely match moments central to our exercise. For instance, while most moments are given a weight of one, we increase the weights on the returns to experience and tenure (the estimated coefficients $\{\hat{\xi}_k, \hat{\zeta}_k\}$ in Equation 14) by a factor of three. We also scale each moment by the square of its value computed from the data.[36]

[35] Yamaguchi (2010) uses a similar strategy in a related model.

[36] Equivalently, we express the distance between a simulated moment and its data value as the deviation rate from its data value.

Where we control for unobserved firm heterogeneity in the real data, we explicitly control for the state variable θ representing the specific component of productivity. In practice, given that θ is discretized on a grid, we include dummies for each grid point as controls.

Our optimization procedure follows closely the "TikTak" optimization algorithm described in Arnoud, Guvenen and Kleineberg (2019). The TikTak algorithm is a global optimization method that combines a grid search step with a local optimization step, and it fits the moment targets better than alternative global optimization methods (e.g., multi-level single-linkage) in our setting. In practice, we perform a grid search over 10,000 parameter vectors and select the 100 vectors giving the best fit as starting points for the local optimization step. We use the NLOPT implementation of the Subplex algorithm as the local optimization method (Rowan, 1990).

C Restricted models

In addition to the baseline model with all four state variables (θ, s, g, and ε), we also estimate several restricted versions of the model. Starting from a model with a single match-specific shock (ε in our notation), we incrementally switch on permanent firm heterogeneity (ε and θ, essentially the model in Krolikowski (2017) with a different bargaining protocol), and general human capital (ε, θ and g). For completeness, the calibrated parameters of each restricted model are reported in Table 3. Importantly, the parameters of each restricted model are obtained from a full calibration to the exact same moment targets as the baseline model, so that each model can, in theory, match the data. The parameters of the restricted models therefore differ from those of the baseline model since they reflect how the optimizer balances the fit of each moment given the parameter restrictions.

D Identification

We provide evidence on the identification of key parameters using two complementary approaches. The first is global and relies on the restricted models described in Section 3.5 and Appendix C. The second is local and consists in computing the Jacobian of the moment vector with respect to the parameters around their calibrated values.

Global identification Figure 6 uses the restricted models to illustrate the identification of the skill accumulation parameters. The left panel shows the returns to experience across models. Models without general human capital accumulation (ε only, and ε and θ) systematically under-predict the returns to experience, suggesting that this moment primarily identifies

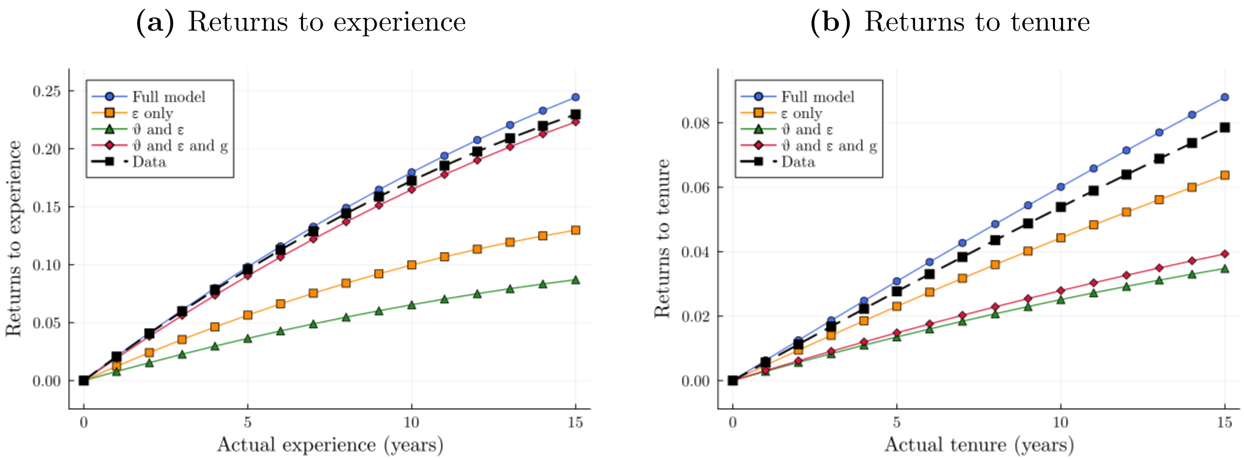
Table 3: Calibrated parameters of restricted models

Parameter	Description	Full model	Restricted models		
			ε only	θ and ε	θ, ε and g
σ_θ	Firm type: $\theta \sim \ln \mathcal{N}(0, \sigma_\theta)$	0.057	—	0.211	0.167
ρ_ε	Persistence AR(1) for match-specific shocks (13)	0.843	0.557	0.393	0.860
σ_ε	Standard dev. AR(1) for match-specific shocks (13)	0.082	0.544	0.283	0.173
λ_N	Contact rate non-employment	0.189	0.013	0.244	0.199
λ_E	Contact rate employment	0.089	0.012	0.276	0.149
δ	Exogenous job destruction rate	0.003	0.008	0.003	0.006
$\ln s_{n_s}$	Max level of firm-specific human capital	0.218	—	—	—
$\ln g_{n_g}$	Max level of general human capital	0.362	—	—	0.330
γ	Appreciation rate firm-specific human capital	0.014	—	—	—
ϕ_E	Appreciation rate general human capital	0.030	—	—	0.034
ϕ_N	Depreciation rate general human capital	0.044	—	—	0.061
α	Worker bargaining weight	0.829	0.626	0.359	0.533
b	Home production factor: $z(g) = b \cdot g$	1.207	0.563	1.327	1.467
λ_r	Reallocation shock rate (conditional on δ-shock)	0.345	0.599	0.476	0.573

Notes: The table shows the calibrated parameters in each version of the model. The third column shows the parameters of the baseline model with all four state variables (θ, s, g, and ε). The fourth column shows the parameters of a restricted model with only match-specific shocks (ε only). The fifth column shows the parameters of a restricted model with match-specific shocks and permanent firm heterogeneity (ε and θ). The sixth column shows the parameters of a restricted model with match-specific shocks, permanent firm heterogeneity, and general human capital (ε, θ, and g). All parameters are obtained from a full calibration to the same moment targets as the baseline model. "—" denotes parameters set to zero in the corresponding model.

the general human capital parameters (γ_g, ϕ_E). The right panel shows the returns to tenure across models. Models that additionally shut down specific human capital accumulation (ε and θ, and ε and θ and g) under-predict the returns to tenure, pointing to these returns as the main source of identification for the specific human capital accumulation rate γ.

Figure 6: Identification of skill accumulation parameters



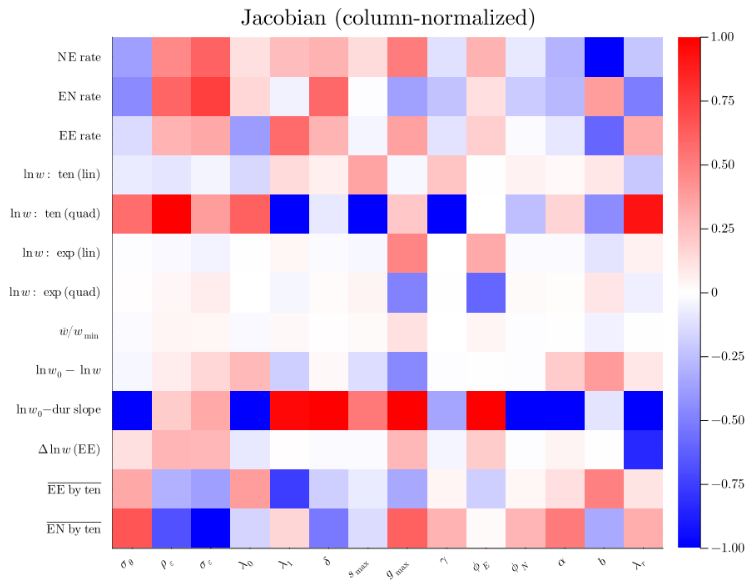
Notes: Each panel shows the fit of the baseline model and the three restricted models to the returns to experience (left) and the returns to tenure (right) estimated from Equation (14). The parameters of each restricted model are obtained from a full calibration to the same moment targets as the baseline model.

Local identification Figure 7 displays the Jacobian matrix of the moment vector with respect to the model parameters, evaluated at the calibrated values. We estimate the Jacobian by simulating the model on a parameter-specific grid around the calibrated values,[37] fitting a fourth-order polynomial to the simulated moments, and taking its derivative at the calibrated values. We normalize each column to lie in $[-1, 1]$, so the plot shows the relative sensitivity of each moment to each parameter. The separation-by-tenure profiles are averaged across tenure bins and labelled $\overline{\text{EE by ten}}$ and $\overline{\text{EN by ten}}$.

The Jacobian broadly confirms the heuristic identification mapping provided in the main text. For example, the skill depreciation rate ϕ_N is strongly associated with the slope of re-employment wages with respect to nonemployment duration, the job destruction rate δ comoves primarily with the EN rate, and the parameters of the idiosyncratic productivity process $(\rho_\varepsilon, \sigma_\varepsilon)$ load heavily on the EN separation profile by tenure. As with any medium-scale structural model, the parameter-moment mapping is not one-to-one, and some parameters influence multiple moments simultaneously.

[37]The grid spans ±50% of the calibrated value for most parameters, and ±80–90% for parameters bounded away from zero (λ_1, δ, γ, ϕ_E, ϕ_N). In all cases the range is clipped to the parameter bounds. Each univariate grid uses 51 equally spaced points.

Figure 7: Jacobian of moments with respect to parameters



Notes: The figure shows the column-normalized Jacobian matrix of the moment vector with respect to the model parameters, evaluated at the calibrated values. Each entry shows the derivative of a moment (y-axis) with respect to a parameter (x-axis), normalized so that each column lies in $[-1, 1]$. The Jacobian is estimated by simulating the model on a parameter-specific grid around the calibrated values, fitting a fourth-order polynomial, and evaluating its derivative at the calibrated values.

E Additional moments

Figure 8 shows the fit of the model to additional moments that are not targeted in the main calibration procedure. These moments include:

- The unconditional EN and EE separation rates by monthly tenure (figures 8a and 8b), relative to the first month. The baseline calibration targets the relative EN and EE separation rates by yearly tenure (figures 1a and 1b). We find that the model replicates the monthly profiles in the first two years of tenure reasonably well. The data shows some spikes at specific tenure durations. These spikes are likely associated with fixed-term contracts. The model does not have a notion of fixed-term contracts, and therefore cannot capture this feature of the data.

- Nonemployment duration (Figure 8c). The figure shows the distribution of nonemployment duration in the data and in the model. The model tends to under-predict longer nonemployment spells. However, looking the cumulative distribution, most workers are back in employment within three years both in the model (approx. 95%) and in the data (approx. 80%).

- Re-employment wage by nonemployment duration (Figure 8d). As an alternative to the auxiliary model given by Equation 15, which imposes a linear relationship between nonemployment duration and re-employment wages, this figure shows that the model captures reasonably well non-linearities including indicators for yearly nonemployment duration.

- Wage growth of stayers and EE movers (Figures 8e and 8f). The model captures the monthly wage growth of stayers very well, and notably the large mass of workers with approximately zero wage growth (Figure 8e). It does less well in capturing the wage growth of EE movers. Any search model with a wage ladder implies a wage-growth distribution that is skewed toward positive wage changes, and this is also a property of the model developed in this paper. Still, the model generates a non-negligible probability of negative wage growth through the bargaining mechanism and through the relocation shock λ_R, but this is not as pronounced as in the data.

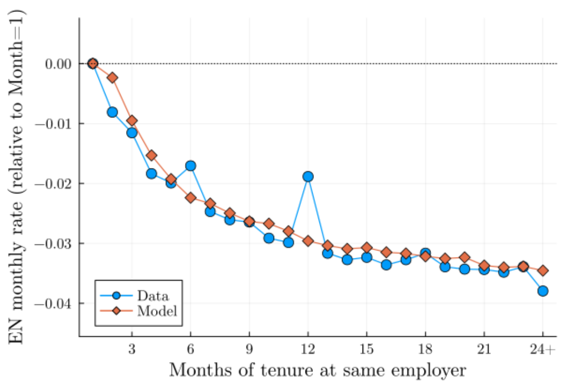
F Sequential auction vs Nash-bargaining

The baseline model uses the sequential auction wage-setting protocol described in (Postel-Vinay and Turon, 2010). This Appendix considers Nash-bargaining as an alternative wage-determination mechanism. Keeping all model parameters at their baseline value, we instead assume that workers' value is given by a constant fraction α of the aggregate job surplus $W(\theta, \varepsilon, s, g) = U(g) + \max\{0, \alpha S(\theta, \varepsilon, s, g)\}$ and invert the value function to obtain the corresponding Nash-bargaining flow wage.

As a simple assessment of the wage dynamics implied by each protocol, Figure 9 reports the monthly wage growth distribution of stayers (panel a) and movers (panel b). Only the sequential auction wage-setting protocol can replicate the large fraction of stayers with no or marginal wage growth (around 90%) in any given month found in the data. With Nash-bargaining, by contrast, this distribution is markedly more dispersed with the fraction of stayers with no wage change dropping to around 10%. Both wage protocols tend to generate excess positive wage growth for movers compared to the data, although the overall dispersion implied by the Nash-bargaining protocol is closer. Overall, this exercise suggests the distribution of stayers' wage growth is helpful in distinguishing between these two wage-setting protocols within this class of models.

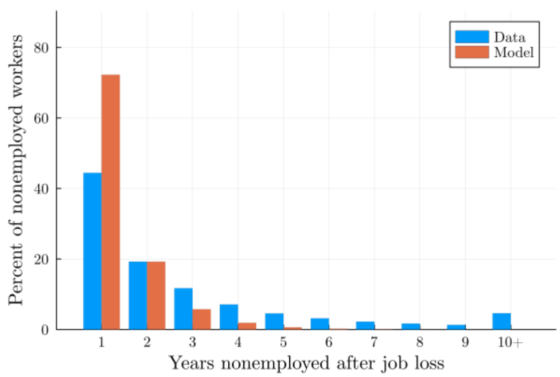
Figure 8: Fit of additional moments

(a) Monthly EN-tenure profile

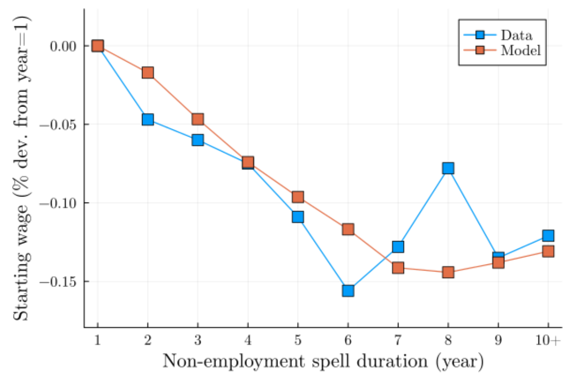


(b) Monthly EE-tenure profile

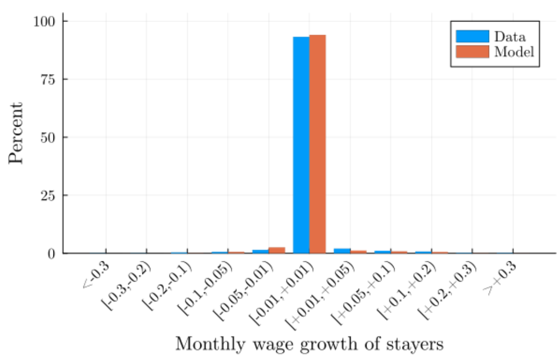
(c) Nonemployment duration



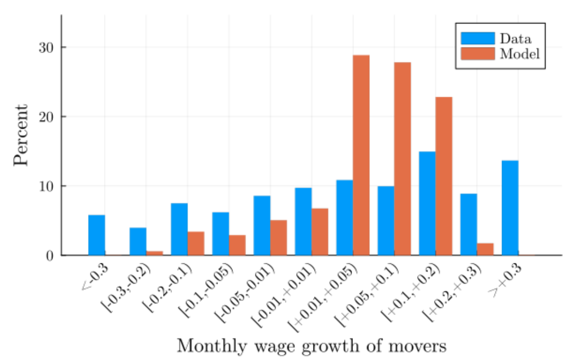
(d) Re-employment wage by duration



(e) Wage growth of stayers

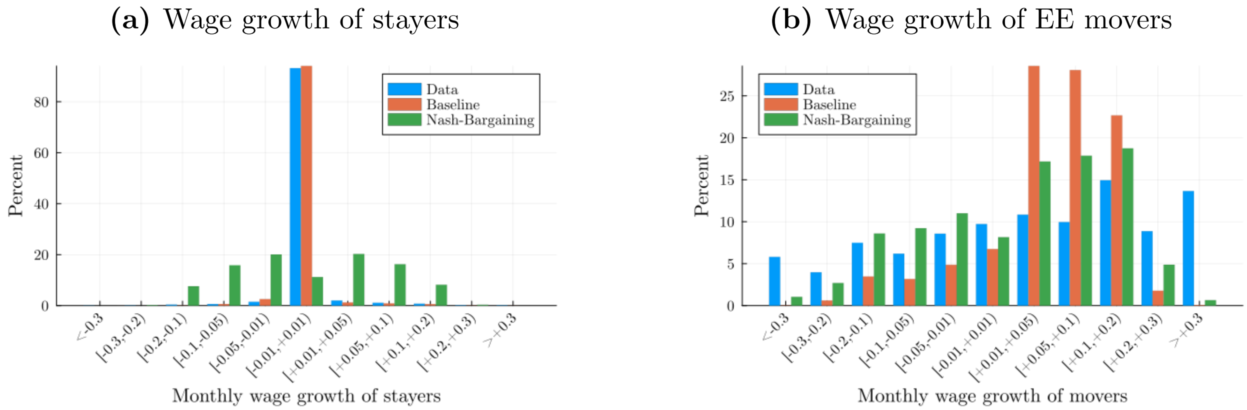


(f) Wage growth of EE movers



Notes: The figure shows the fit of the model to additional moments not targeted in the main calibration procedure. Panel (a) shows the relative EN separation rate within the first two years of a worker's tenure. Panel (b) shows the relative EE separation rate within the first two years of a worker's tenure. Panel (c) shows the distribution of nonemployment duration. Panel (d) shows workers' relative re-employment wages by nonemployment duration. Panel (e) shows the wage growth of stayers, and Panel (f) shows the wage growth of EE movers. See main text for additional details.

Figure 9: Wage growth: baseline vs. Nash-bargaining



Notes: The figure compares the fit of the baseline model and the model with Nash-bargaining to the corresponding wage growth distributions in the data. Panel (a) shows the wage growth of stayers, and Panel (b) shows the wage growth of EE movers. See main text for additional details.

G Robustness of structural decomposition of losses

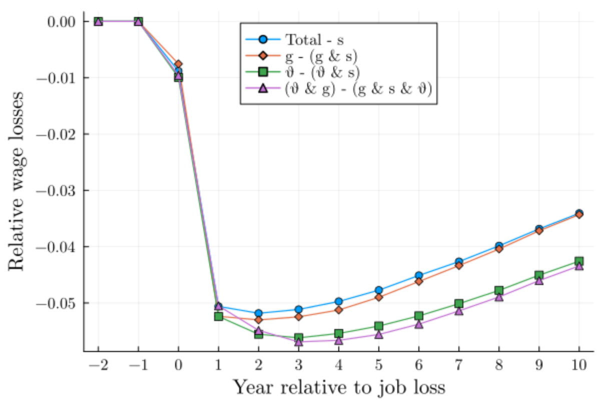
To assess how the order in which the counterfactual losses are constructed affects the overall decomposition, we try out various permutations of the state variables $\{\theta, s, g\}$. Using specific human capital (s) as an example, the contribution of s to the overall losses can be represented in four different ways: (i) the difference between the losses in the treated group and the losses in the counterfactual group with s of the controls "Total - s"; (ii) the difference between the losses in the counterfactual group with g of the controls and the losses in the counterfactual group with g and s of the controls "$s - (s+g)$"; (iii) the difference between the losses in the counterfactual group with θ of the controls and the losses in the counterfactual group with θ and s of the controls "$\theta - (\theta + s)$"; and (iv) the difference between the losses in the counterfactual group with g and θ of the controls and the losses in the counterfactual group with s and g and θ of the controls "$(\theta + g) - (g + s + \theta)$".

Figure 10a shows the four corresponding series for specific skills. As described in the main text, the contribution of s to the overall losses is larger in the counterfactuals where the treated group is assigned the s of the controls *after* being assigned the θ of the control group. This is the case for counterfactuals "$\theta - (\theta + s)$" and "$(\theta + g) - (g + s + \theta)$." Figures 10b and 10c report the results of a similar exercise, respectively, for firm productivity (θ) and general human capital (g). The pattern for firm productivity (Figure 10b) mirrors that for specific skills.

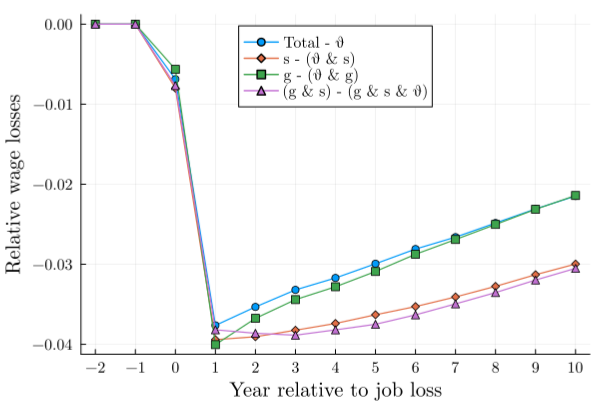
The main message from this exercise is that the order in which specific human capital (s) and firm permanent productivity (θ) are switched on significantly affects their respective contribution to total wage losses. By contrast, the order in which general human capital (g) is switched on makes little difference. Intuitively, a high-level of specific human capital is more valuable at a relatively high-θ firm. At a low-θ firm, workers find it optimal to switch jobs again even with a high level of specific human capital, and these specific skills are lost following such a job-to-job transition. As a result, the contribution of s is larger in the counterfactual decomposition where the treated are assigned the s of the control group *after* being assigned the θ of the control group. This mechanism is not at play with general human capital (g), which is fully transferable.

Figure 10: Alternative order in wage decomposition

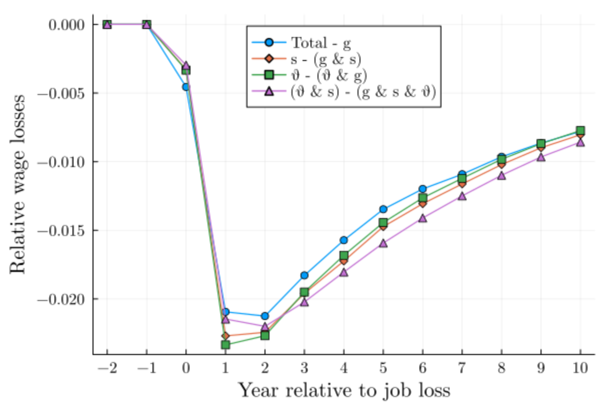
(a) specific skills (s)



(b) Employer productivity (θ)



(c) General skills (g)



Notes: The figure shows the contribution to wage losses of specific human capital (panel a), employer productivity (panel b), and general human capital (panel c). Each line corresponds to different assumptions on the construction of the counterfactual decomposition. See Appendix G for details.